

Erica C.N. Robertson

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Education

- May 2022 – Present **Colorado State University**
PhD in Ecology through the Graduate Degree Program in Ecology, housed in the Department of Biology
Expected Defense: March 2027
- Aug 2016 – May 2020 **George Washington University**
Bachelor of Science in Molecular and Cellular Biology
- Jan – May 2019 **Universidad San Francisco de Quito**
IES exchange program at the Galapagos Academic Institute for the Arts and Sciences
- Aug 2015 – May 2016 **Parkland College**

Research Experience

- Aug 2022 – Present **PhD Student in the Kristen Ruegg Lab, Colorado State University**
- Integrate population genomics, morphology, ecological data, and climate projections to study adaptation and conservation risk in North American birds.
 - Identified genomic and morphological signatures of local adaptation in Gray-crowned Rosy-Finches across contrasting alpine environments.
 - Use historical and contemporary whole-genome data to evaluate temporal changes in genetic diversity, demography, and climate-associated variation in Brown-capped Rosy-Finches.
 - Defined genomic conservation units in Bank Swallows and integrated these with eBird Status and Trends data to evaluate environmental drivers of population decline and future climate vulnerability.
 - Collaborate with museums, state and federal agencies, international researchers, and conservation practitioners to translate genomic results into conservation recommendations.
 - Contribute to Bird Genoscape Project research and broader efforts to apply genomic data to avian conservation.
- Aug 2019 – July 2020 **Research Assistant in the Arnaud Martin Lab, The George Washington University**
- Designed and carried out experiments using CRISPR/Cas9 to perform gene knockouts
 - Reared butterflies and imaged wing patterns
 - Collected sequence data from NCBI and generated phylogenetic trees
- June – Aug 2018;
Jan – May 2021 **Research Assistant in the Mark Davis Lab, University of Illinois**
- Performed qPCR analysis of environmental DNA (eDNA) samples targeting Odonata
 - Tested PCR primers for cicada microsatellite analysis
 - Completed and trained students in phenol-chloroform extractions of eDNA samples
 - Executed RADSeq digestions and adaptor ligation in preparation for sequencing
 - Trained and oversaw lab techs in various procedures such as PCR prep and gel electrophoresis
- June – Aug 2018 **Research Assistant in Chris Dietrich Lab, University of Illinois**
- Organized and cataloged hundreds of Membracoidae specimens

- Performed dissections and Qiagen DNA extractions on Cicadellidae specimens in preparation for sequencing

June – Aug 2018

Research Assistant in Hugh Robertson Lab, University of Illinois

- Manually annotated and analyzed Odorant Binding Protein genes in aphid genomes using the Apollo Genome Browser available at AphidBase

Publications

- Robertson, E. C. N.**, et al (2026) The Genetic and Morphological Basis of Local Adaptation to Elevational Extremes in an Alpine Finch. *Ecology and Evolution*. <https://doi.org/10.1002/ece3.72962>
- Mastretta-Yanes, A., **et al.** (2024) Multinational evaluation of genetic diversity indicators for the Kunming-Montreal Global Biodiversity Framework. *Ecology Letters*, 27, e14461. <https://doi.org/10.1111/ele.14461>
- Hanly, J., **Robertson, E. C. N.**, et al. (2021). Porcupine/Wntless-dependent trafficking of the conserved WntA ligand in butterflies. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*. <https://doi.org/10.1002/jez.b.23046>
- Robertson, H. M., **Robertson, E. C.**, et al. (2019). The chemoreceptors and odorant binding proteins of the soybean and pea aphids. *Insect Biochemistry and Molecular Biology*, 105, 69-78. <https://doi.org/10.1016/j.ibmb.2019.01.005>

Manuscripts in Review and Preparation

- Robertson, E. C. N.**, et al. (2026) Integrating genomics and hierarchical modeling improves identification of drivers of population decline. *Manuscript in preparation*.
- Forester, B. R., **Robertson, E. C. N.**, DeSaix, M., et al. (2026) A widespread gap in the U.S. Endangered Species Act: Risk of genetic erosion within populations. *In Review at Biological Conservation*. <https://doi.org/10.32942/X2KX06>
- Nikunj, Goel, **et al.** (2026) Identifying adaptive variation in spatially structured populations using low-coverage whole-genome sequencing data. *In Review at Molecular Biology and Evolution*. <https://doi.org/10.64898/2026.01.19.700474>
- Ruegg, K. C., **et al.** (2026) Loss of Adaptive Capacity Drives Climate Vulnerability Across Taxonomic Scales in an Alpine Specialist Species Complex. *In Prep*. <https://doi.org/10.64898/2026.05.13.724772>
- Amirkhiz, R. G., **et al.** (2026) Counterfactual attribution reveals environmental and niche drivers of non-monotonic century-scale species redistribution under climate change. *In Review at Diversity and Distributions*.

Presentations

- Robertson, E. C. N.**, et al (2026) At the Right Scale: Using Evolutionary Significant Units to Diagnose Population Decline in Bank Swallows. *Oral Presentation, American Ornithological Society Annual Meeting, Amherst, MA*.
- Robertson, E. C. N.**, et al (2026) At the Right Scale: Using Evolutionary Significant Units to Diagnose Population Decline in Bank Swallows. *Oral Presentation, Front Range Student Ecology Symposium, Colorado State University, CO.* (Outstanding Oral Presentation)
- Robertson, E. C. N.**, et al (2024) Novel genes underlying locally adapted traits in an alpine finch species. *Oral Presentation, American Ornithological Society Annual Meeting, Estes Park, CO*.
- Robertson, E. C. N.** (2024). Assessing Local Adaptation and Underlying Genes in an Alpine Finch. *Oral Presentation, Front Range Student Ecology Symposium, Colorado State University, CO.* (Outstanding Oral Presentation)

Science Communication and Outreach

- 2027 (scheduled) **Co-leading Road to Recovery Workshop on Genetics**
Providing an introduction on how genetics can be incorporated into conservation action
- 2026 **Guest Speaker for National Math Stars**

	Introducing students, ages 9-11, to the world of ornithology.
2026	Presentation for the San Fernando Valley Audubon Society, Van Nuys, CA Presented on the work done by Bird Genoscape Project and my dissertation work.
2026	Presentation for Denver Audubon, Denver, CO Presented on the work done by Bird Genoscape Project and my dissertation work.
Spring 2025	Genetic Indicators Workshop Developed workshop materials for national workshop on genetic diversity indicators presented to member of the Namibia government and academic groups.
2023	Teacher for Bird Camp An outreach program run by the Bird Genoscape Project that introduces student to birding and ornithology research.

Teaching and Course Design

October 2026 (scheduled)	Guest Speaker for Conservation Genomics, CSU
Spring 2026	ECOL592-The Science and Policy of Species Conservation Codesigned and assisted in running a graduate level course on the integration of science and policy in species listing decision making at the global, federal, and state levels
April 2026	Guest Lecturer for Ornithology, CSU
2025	ECOL592-Nature Journaling Conceptualize, designed, and taught graduate level course on nature journaling
Spring 2022 – Spring 2026	Graduate Teaching Assistant, CSU GTA for LIFE102 (1 semester) and BZ350 (5 semesters)
Aug 2019 – May 2020	Head Learning Assistant, GWU

Service

2024 and 2026	Poster Presentation Judging For CSU's Celebrate Undergraduate Research and Creativity Showcase
2024 and 2025	Undergraduate Research Mentoring As part of CSU's MURALS program and independently through the Ruegg Lab

Scientific Illustration and Visual Communication

Bird Genoscape Project outreach materials.

Designed a public-facing flyer for bird-banding stations explaining how collected feathers contribute to genomic research and the broader Bird Genoscape Project.

Robertson et al. (2026). *Ecology and Evolution*.

Lead figure designer; created all figures and Gray-crowned Rosy-Finch illustrations.

Forester et al. (in review).

Lead figure designer; created all figures, data visualizations, and six organism illustrations.

Durkee et al. (2026). *Molecular Ecology*.

Created scientific illustrations of the Rocky Mountain Clouded Sulfur butterfly for Figure 1.

Zamudio-Beltrán et al. (2025). *Evolution Letters*.

Created Common Yellowthroat illustrations for Figure 3.

Research Grants

2024	Assessing Local Adaptation and Underlying Genes in an Alpine Finch Species, GDPE Research Grant, Colorado State University (\$600)
2024	Assessing Climate Risk in Alpine Finch Species, Lois Webster Grant, Denver Audubon Society (\$3900)

2023 **Assessing Climate Risk in Alpine Finch Species**, Colorado Field Ornithologist (\$850)

Work Experience

Jan 2024 – Current	Graduate Research Assistant, CSU
January – May 2025	Genomics Consultant for the University of Namibia
May 2017 – Dec 2019	Lab Prep Assistant, GWU Biology Department
Sep 2016 – Dec 2017	Explainer, Smithsonian National Air and Space Museum

Honors and Awards

February 2026	Outstanding Oral Presentation The 2026 Front Range Student Ecology Symposium
Fall 2024 and Spring 2025	Sharon E. Kabes Scholarship (\$1200) From Colorado State University
February 2024	Outstanding Oral Presentation The 2024 Front Range Student Ecology Symposium
Fall 202 and Spring 2024	Dr. Heather M. Rueth Ecology Scholarship (\$1900) From Colorado State University
May 2020	Received <i>cum laude</i> honors From George Washington University